FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

标题 Title	GAC T79-G2 HE Engine Thermal shock test

1. Test information

测试项目 Activity	Vibration sweeping	项目状态 Project state	☐ PV ☑ DV ☐ Innovation ☐ Others
客户 Customer	GAC	发动机号: Engine No:	2-5#
测试地点 Location	LVE bench 2	测试日期 Test date	2026.7.31
测试依据 Test method	CHD-I-DSE-2007-Thermal Shock Endurance(1000cycles)		

2. 样件描述 Sample description

样件号 Sample Number	Manifold	1
	Hot End	C004J-1.5T-GPF-9001-DV-002
	Hot End	3
	Bracket	4
	Cold End	C004J-1.5T-CE-9001-DV-001
	Cold End	C004J-1.5T-CE-9001-DV-002
	Cold End	7
	EGR pipe	8
样件检查 Sample inspection		OK

3. 偏差描述 Deviation description

序号 No.	偏差类别Deviation categor	偏差描述 Deviation description	日期 Date
1	Sample deviation	焊接气孔和载体细微磕破，详见就最后一页	7/29/2026
2	Spec. deviation	捅穿GPF做热冲击试验，详见就最后一页	7/31/2026

4. 测量结果 Measurement result

- 振动耐久试验 Vibration endurance test

序号 No.	转速 Speed	跟踪点 Tracking point	阶次 Order	轴向 Axis	循环 Cycles
1					
2					
3					
4					
5					

- 热冲击耐久试验 Thermal endurance test

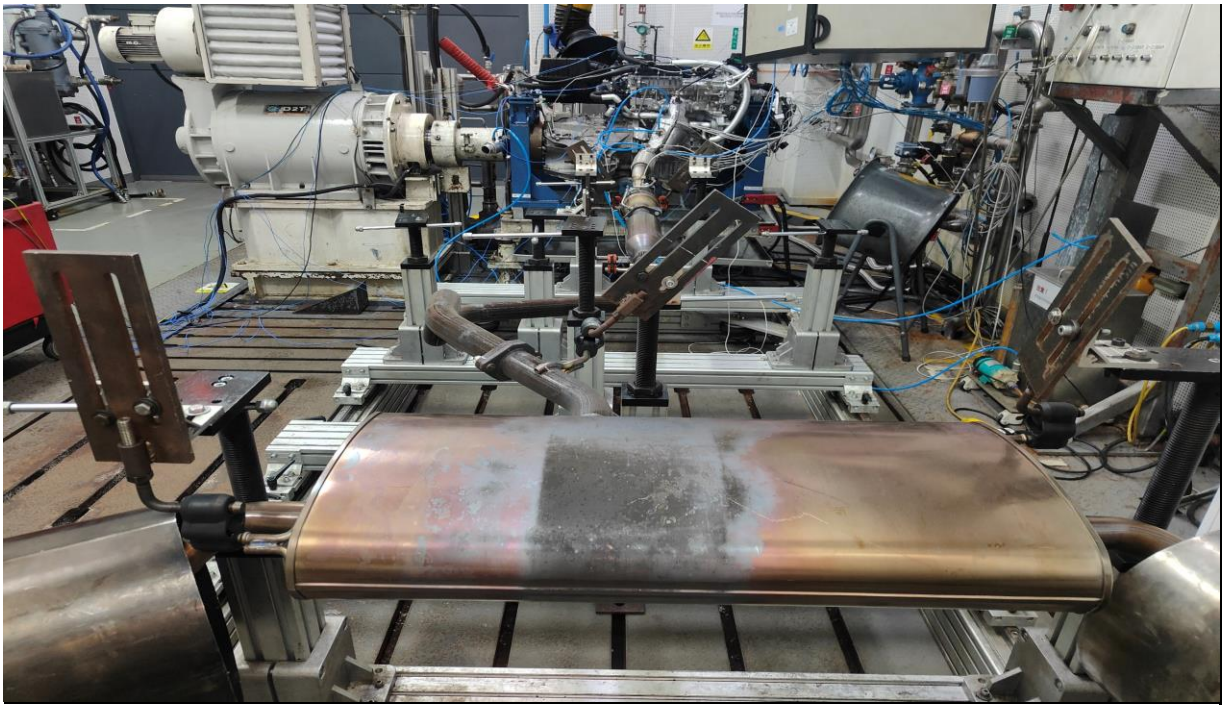
序号 No.	转速 Speed	跟踪点 Tracking point	阶次 Order	轴向 Axis	循环 Cycles
1					

作者 Author	测试工程师 Test engineer	专家 Expert	工程经理 EM
JIA Sibo	ZHAO Wei	QIAN Chris	JIN Xin

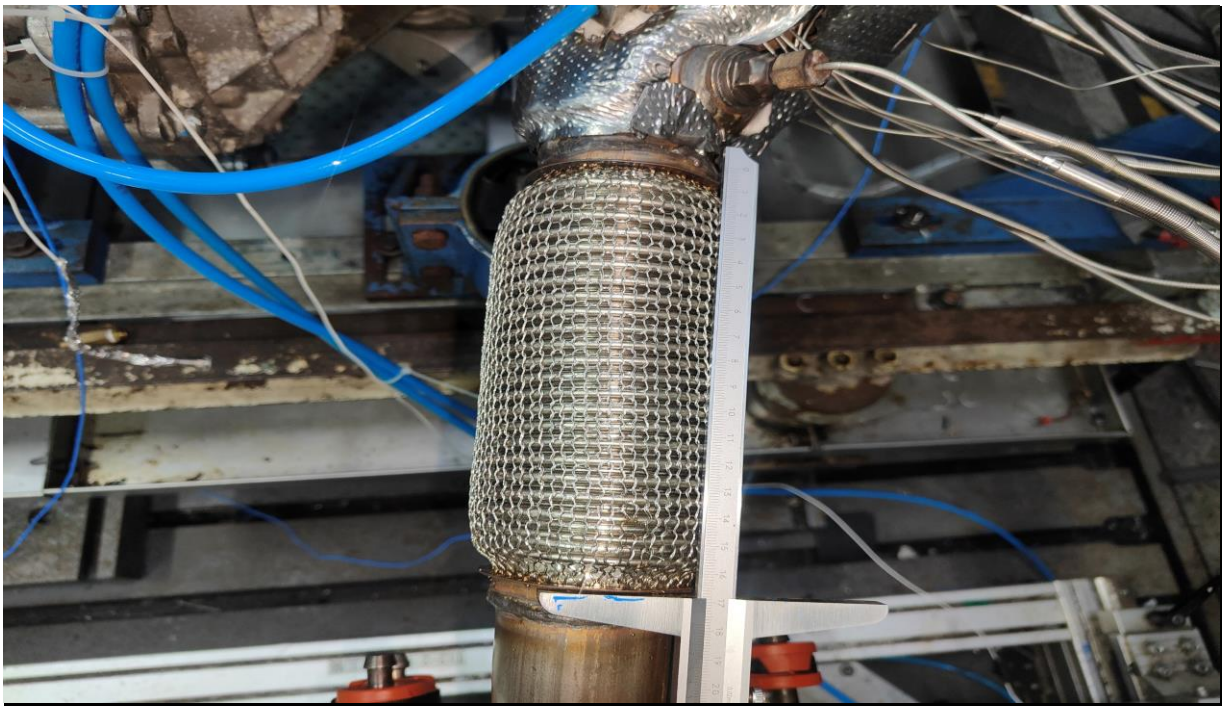
FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

附录 Appendix

1. 试验布置 Test setup



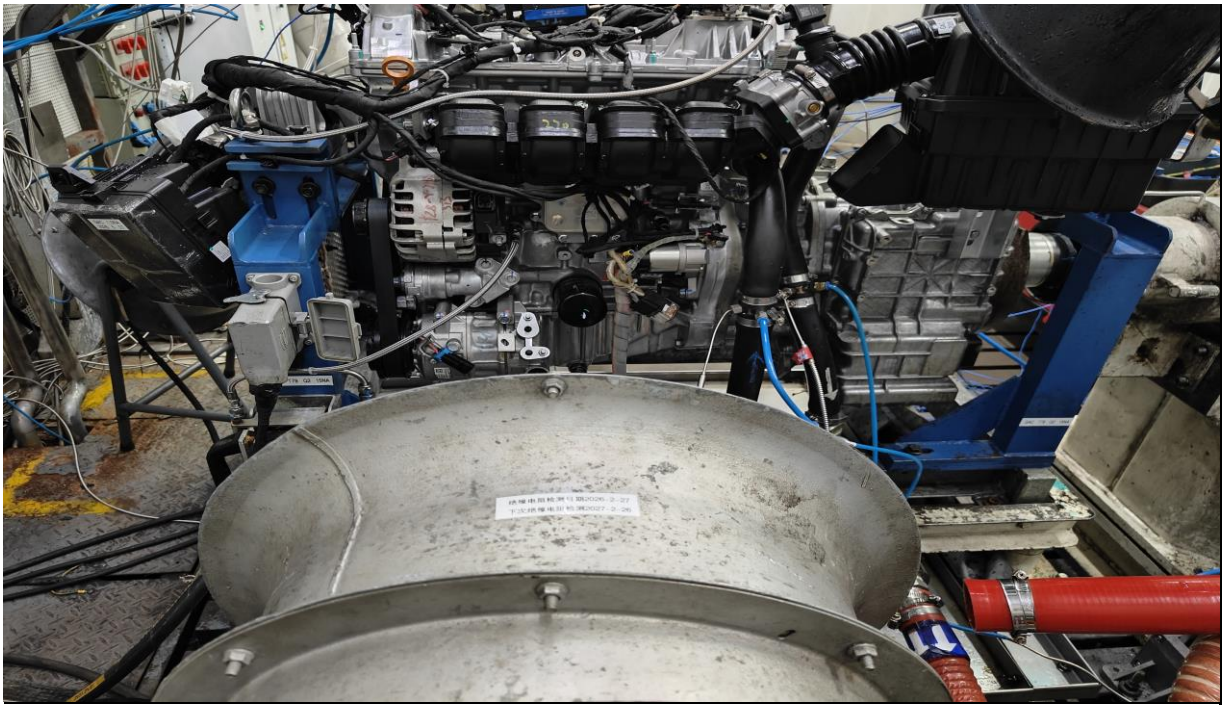
2. 柔性节长度测量和布置 Flex length measure and setup



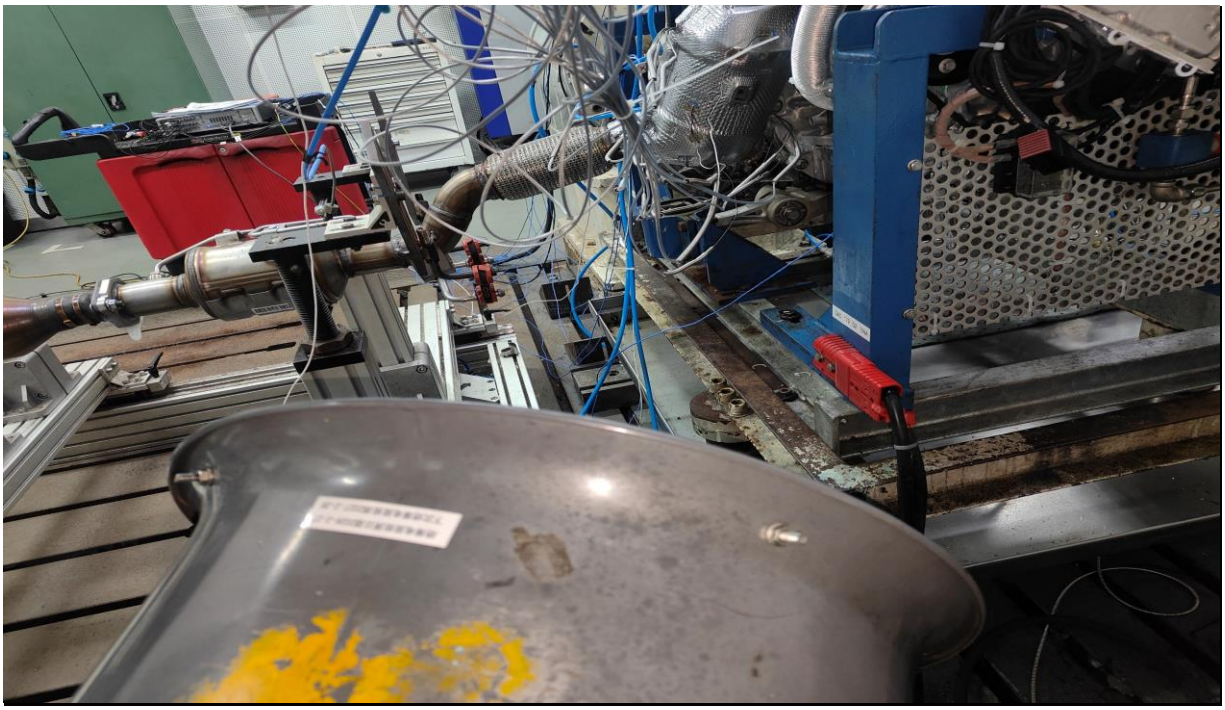
FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

附录 Appendix

3. 发动机风扇布置 Engine fans setup



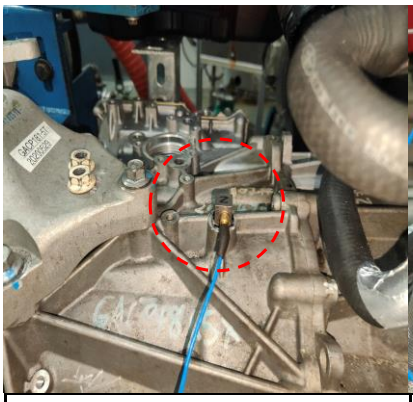
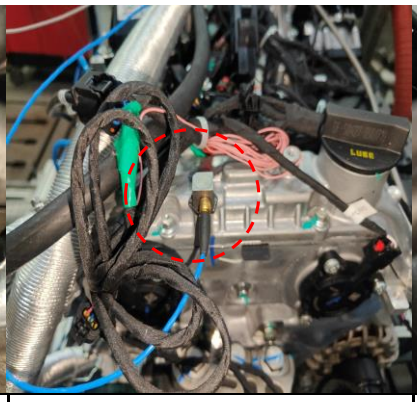
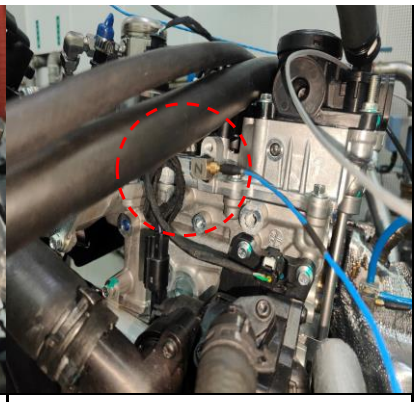
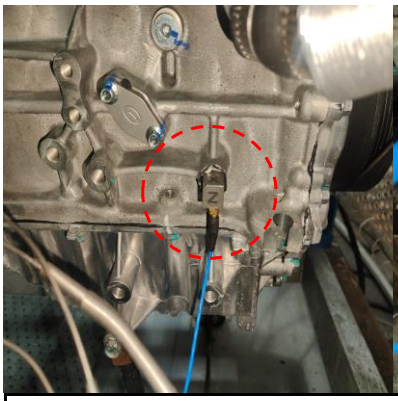
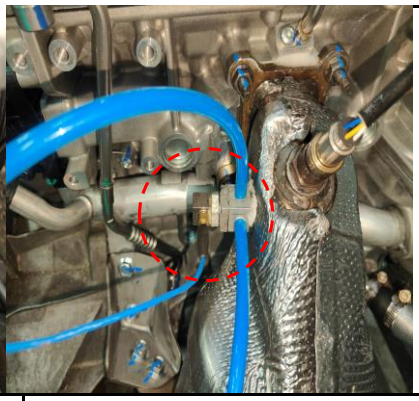
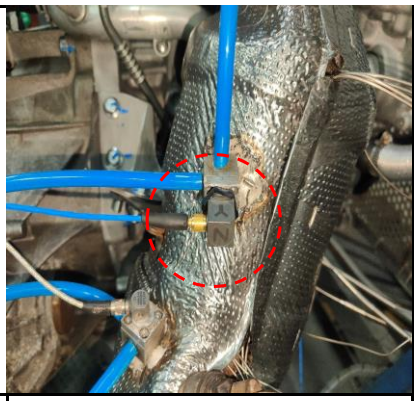
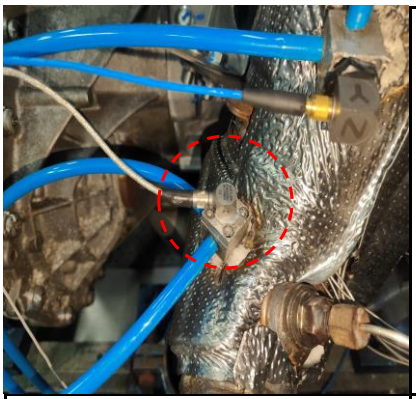
4. 样件风扇布置 Sample fans setup



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附录 Appendix

3. 加速度布置 Acceleration setup

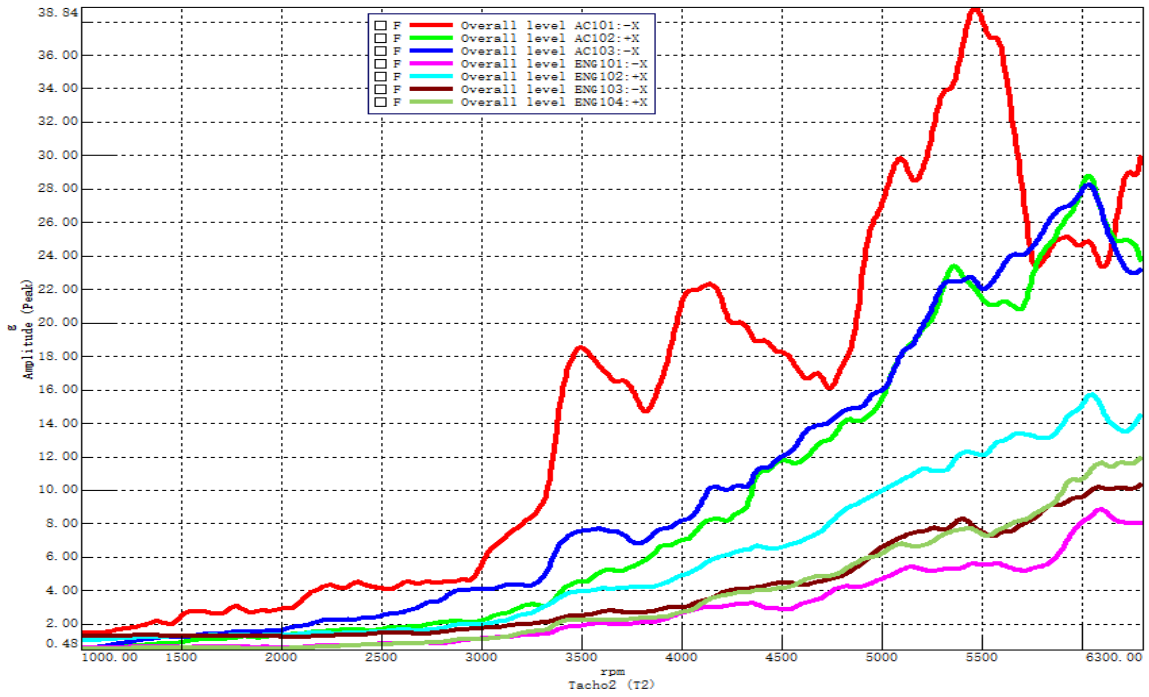
		
ENG101 变速箱	ENG102 缸盖（皮带轮侧）	ENG103 缸盖（变速箱侧）
		
ENG104 缸体	AC101 Inlet cone	AC102 TWC
		
AC103 Outlet cone		

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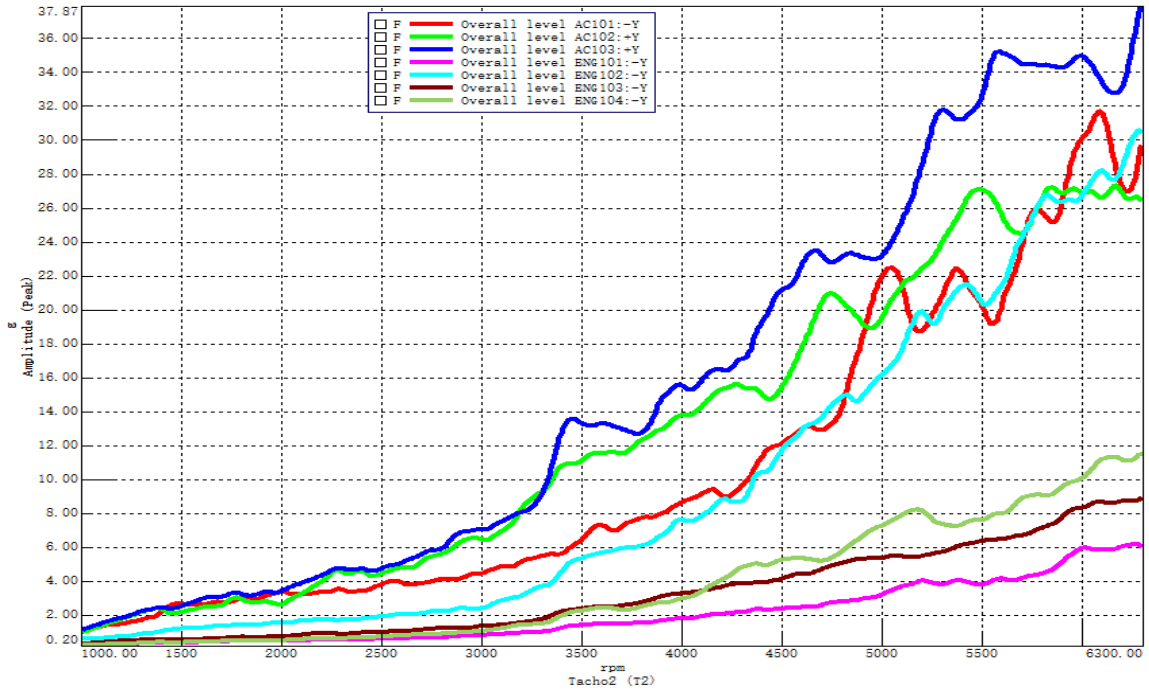
附录 Appendix

4. 振动测量 Vibration measuring

Acceleration_Overall_X_axis

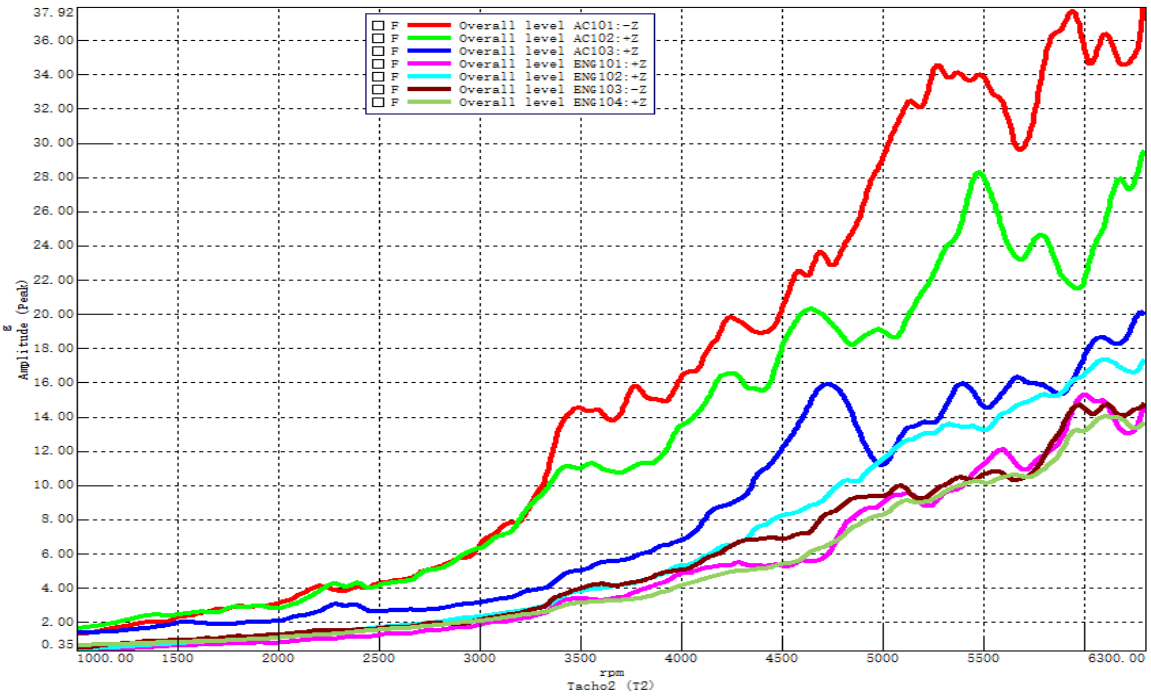


Acceleration_Overall_Y_axis

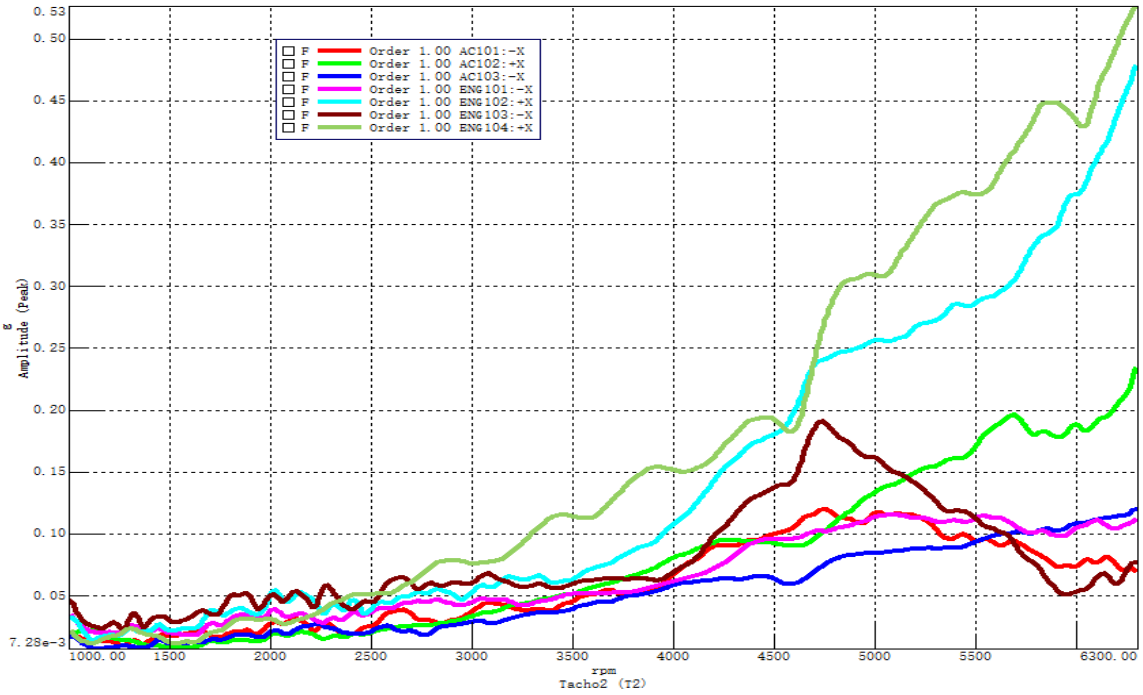


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Acceleration_Overall_Z_axis

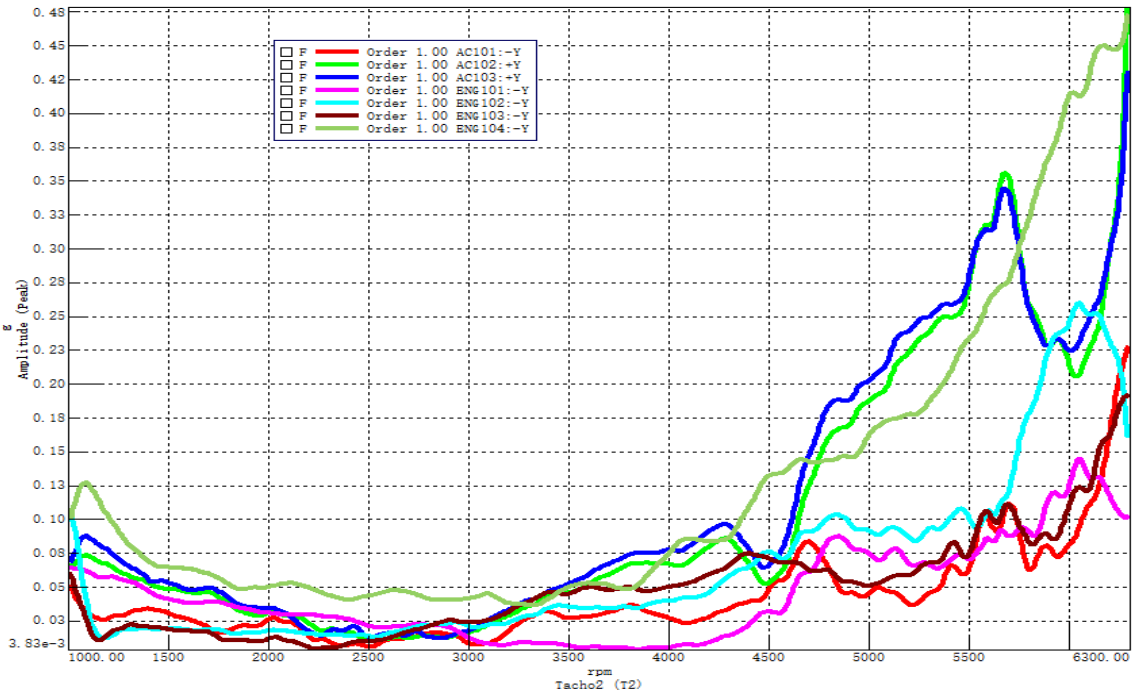


Acceleration_Order_1_X_axis

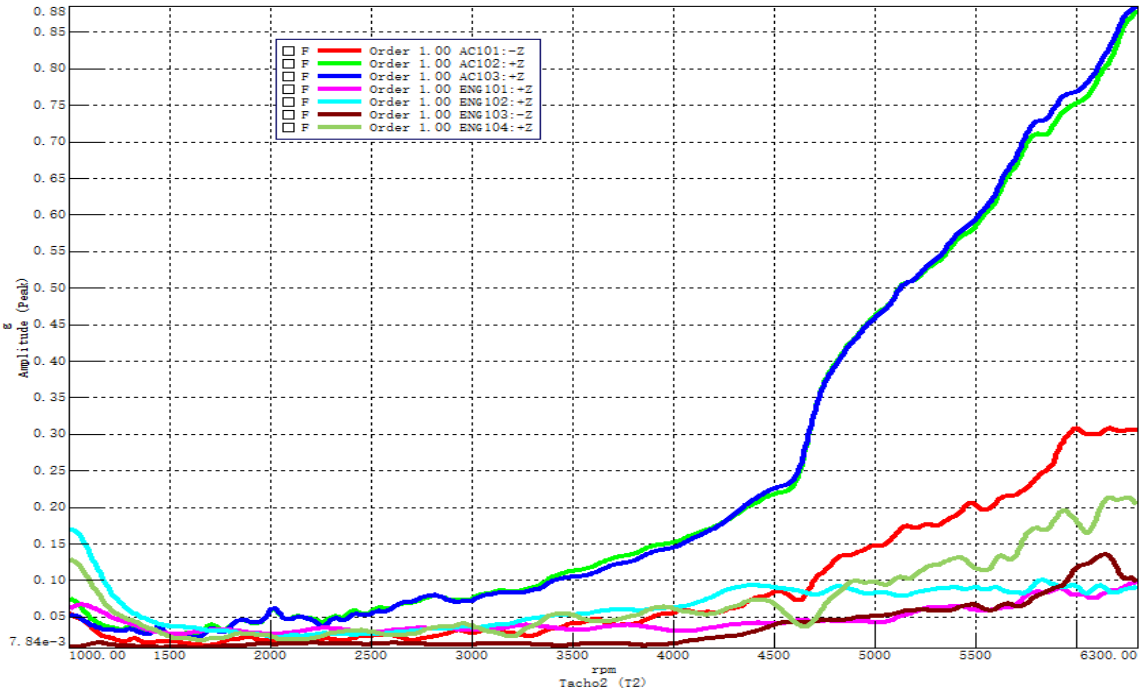


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Acceleration_Order_1_Y_axis

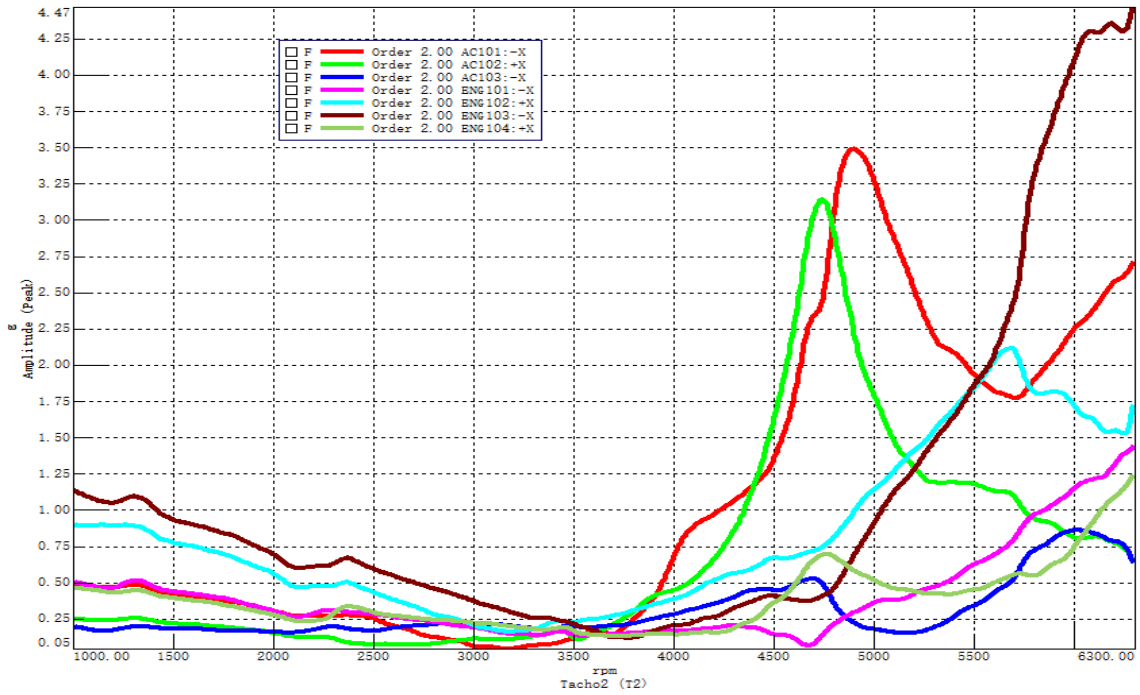


Acceleration_Order_1_Z_axis

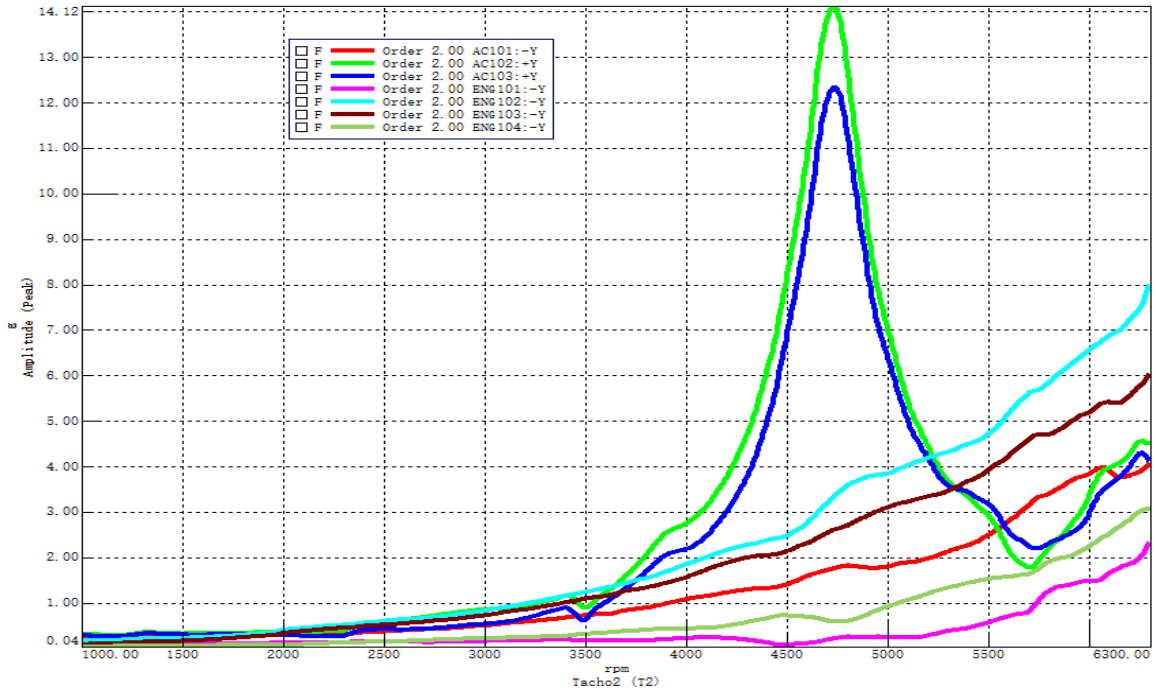


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Acceleration_Order_2_X_axis

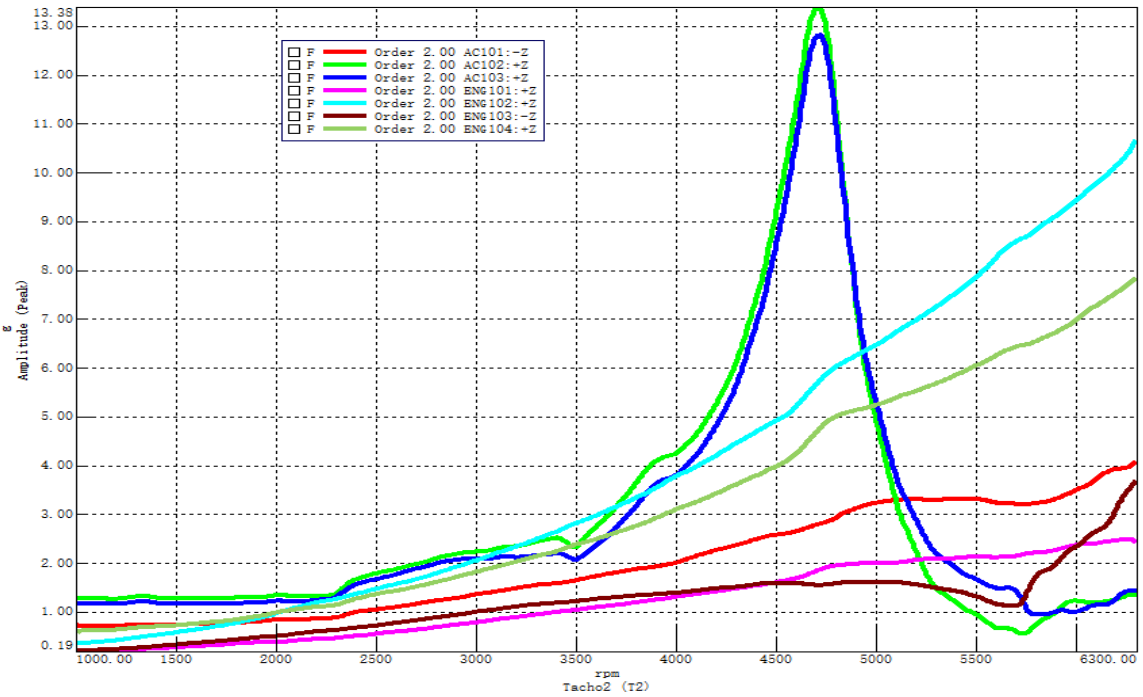


Acceleration_Order_2_Y_axis

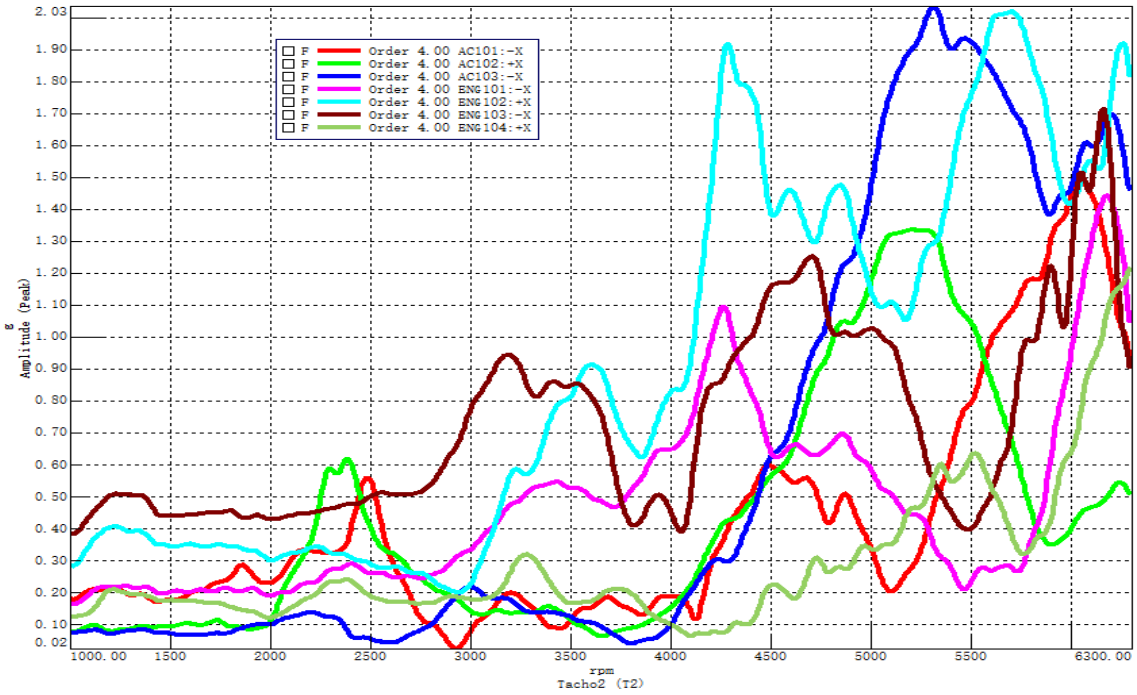


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Acceleration_Order_2_Z_axis

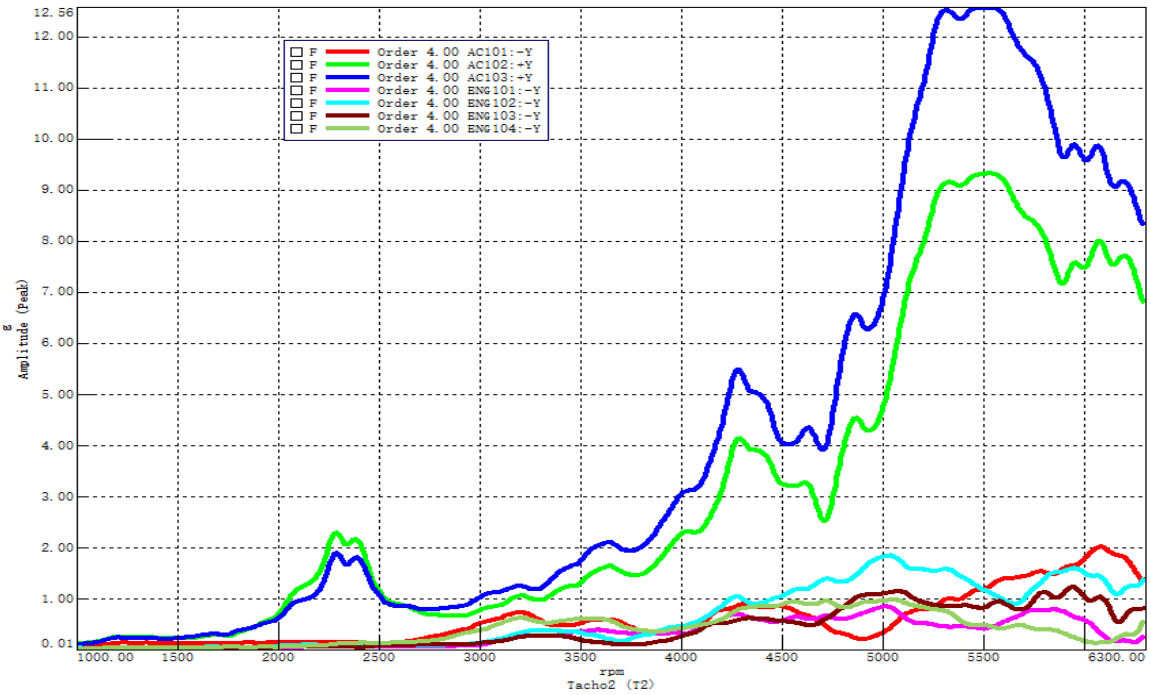


Acceleration_Order_4_X_axis

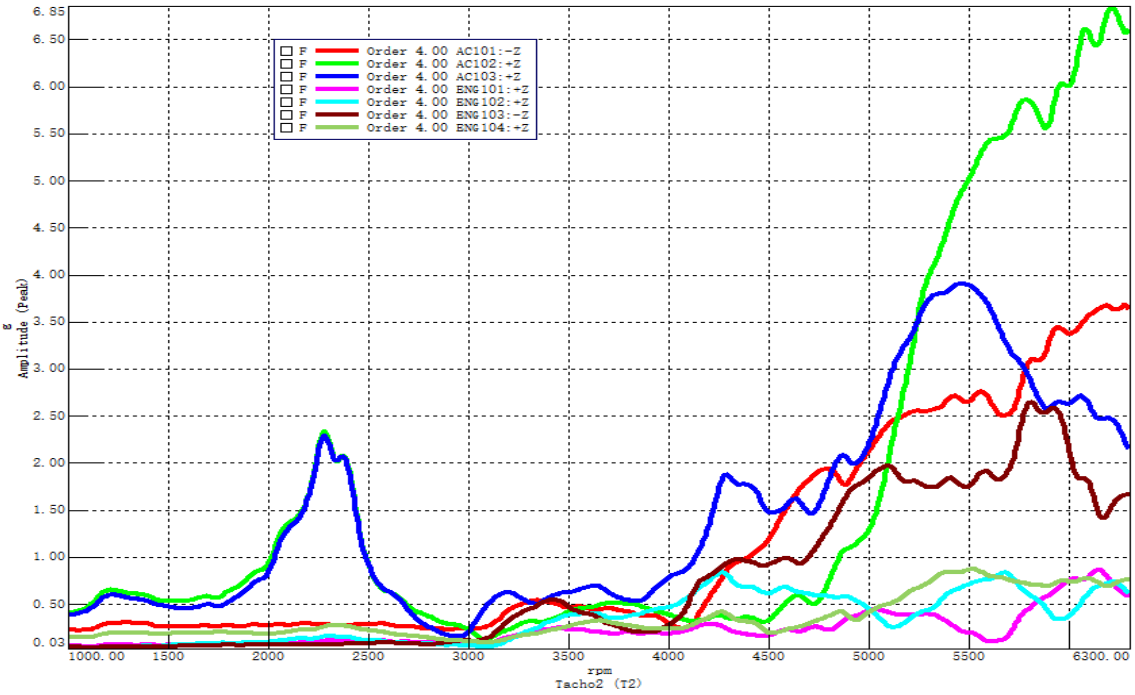


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Acceleration_Order_4_Y_axis

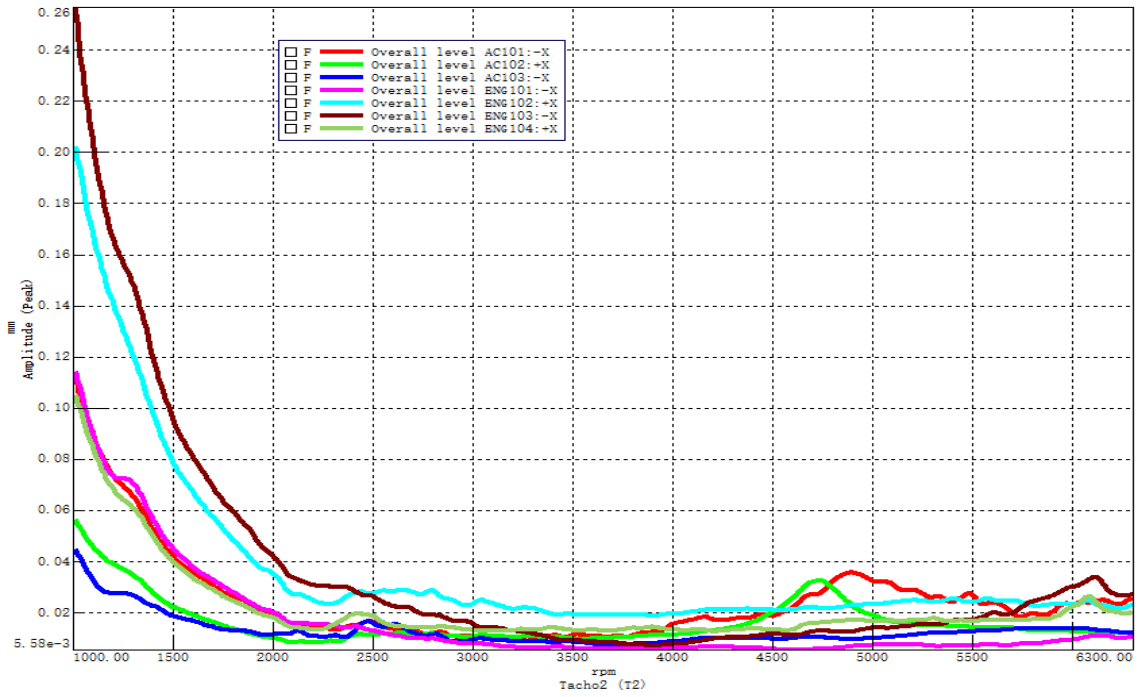


Acceleration_Order_4_Z_axis

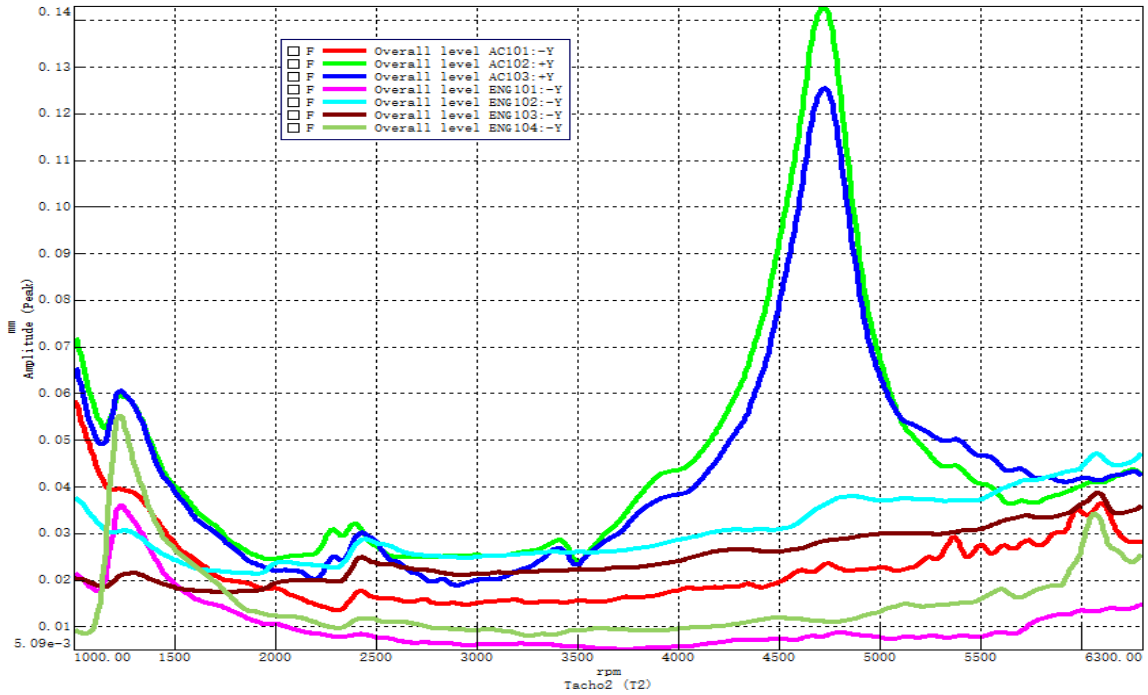


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Displacement_Overall_X_axis

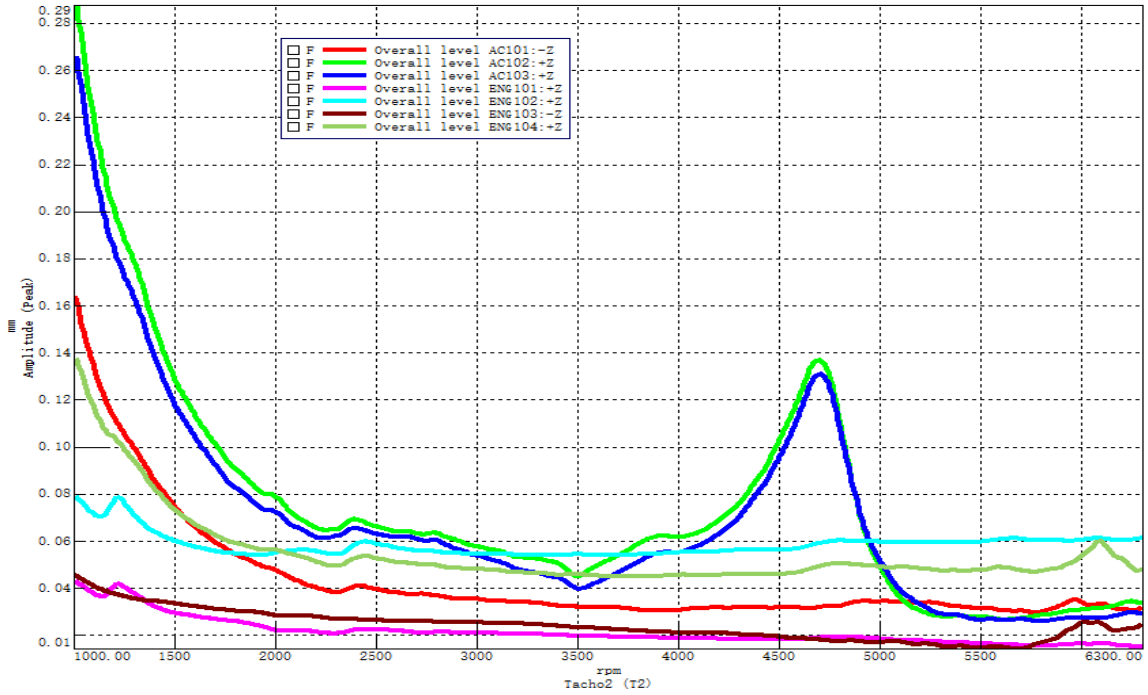


Displacement_Overall_Y_axis

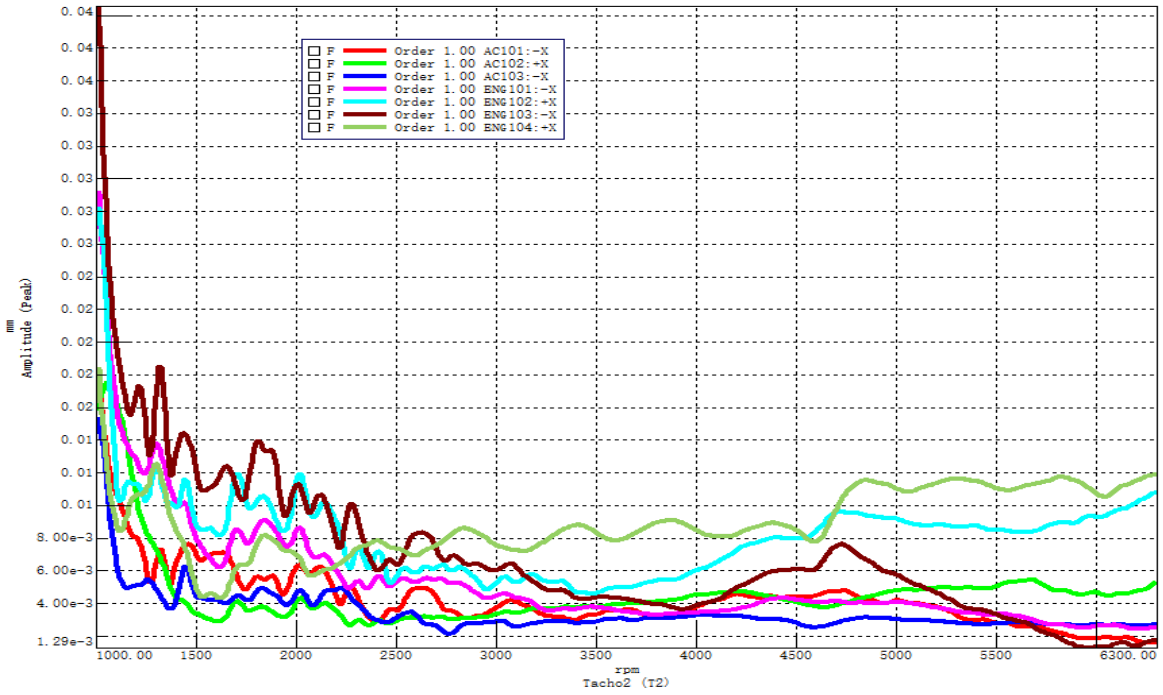


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Displacement_Overall_Z_axis

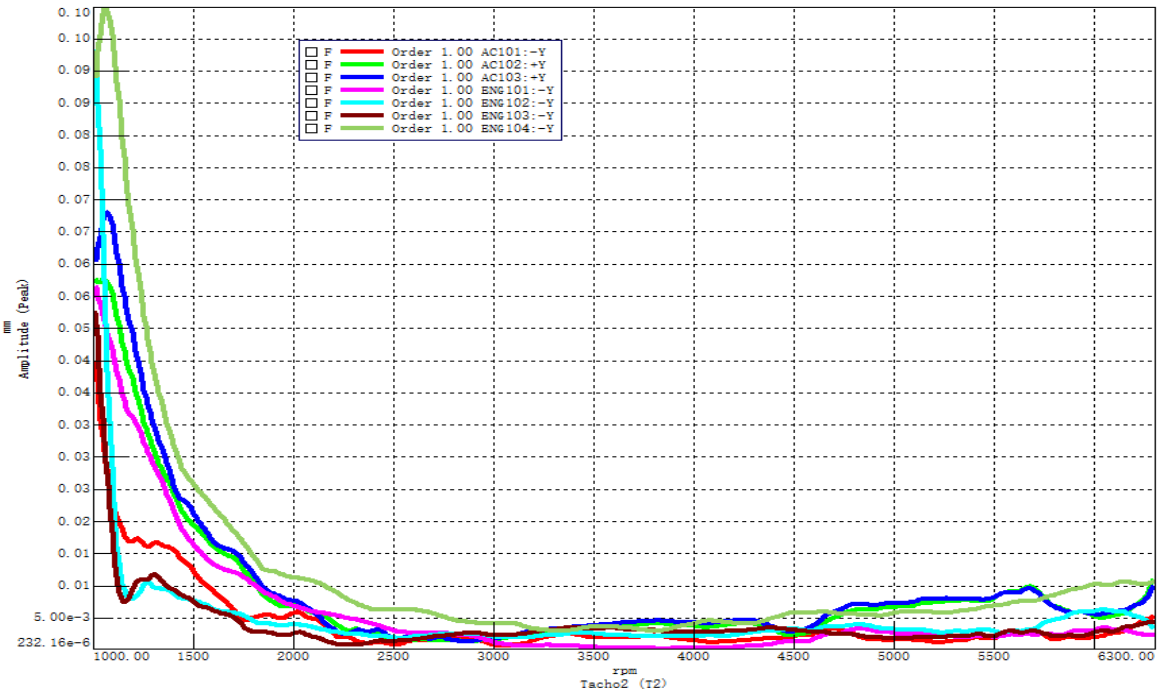


Displacement_Order_1_X_axis

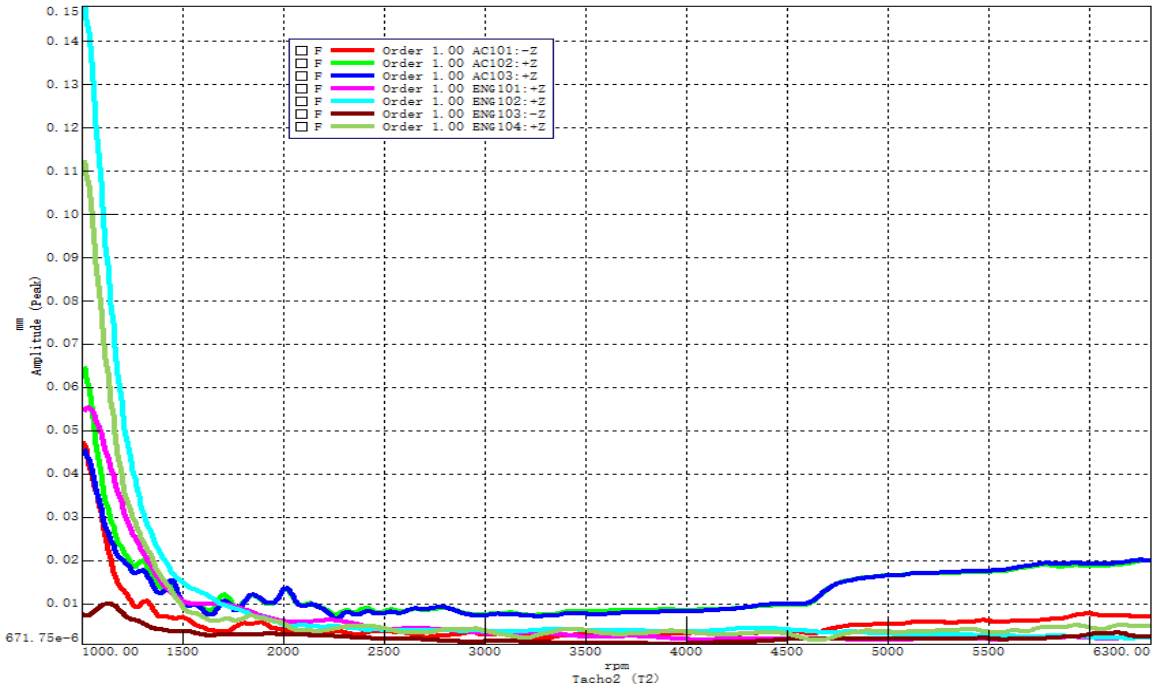


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Displacement_Order_1_Y_axis

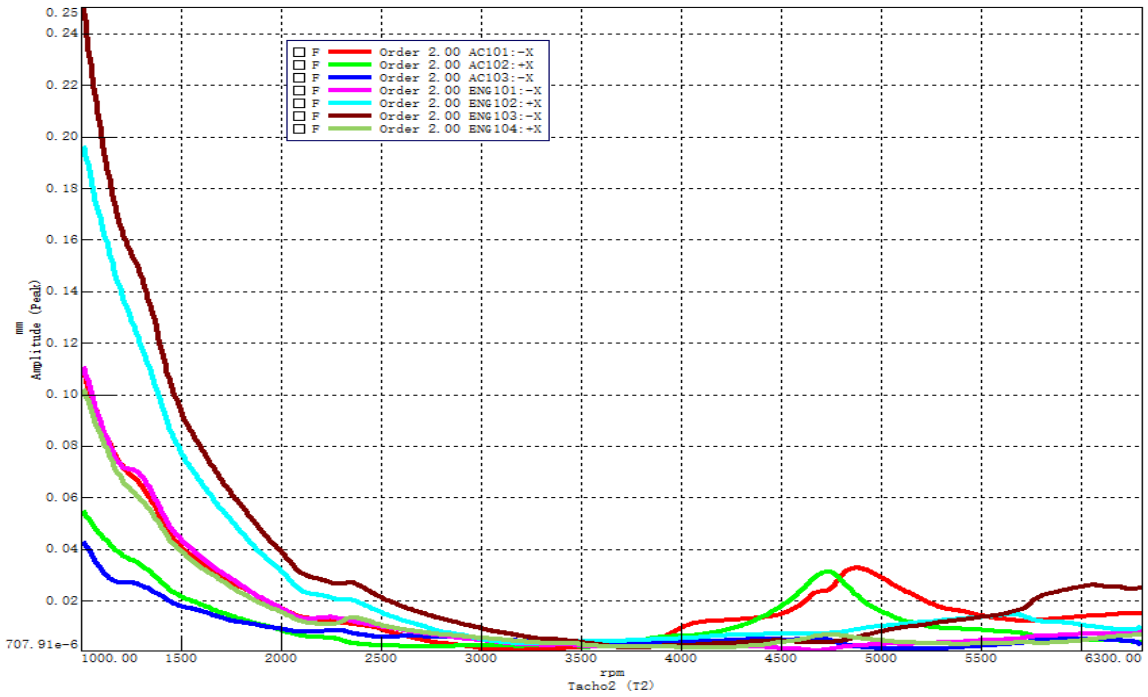


Displacement_Order_1_Z_axis

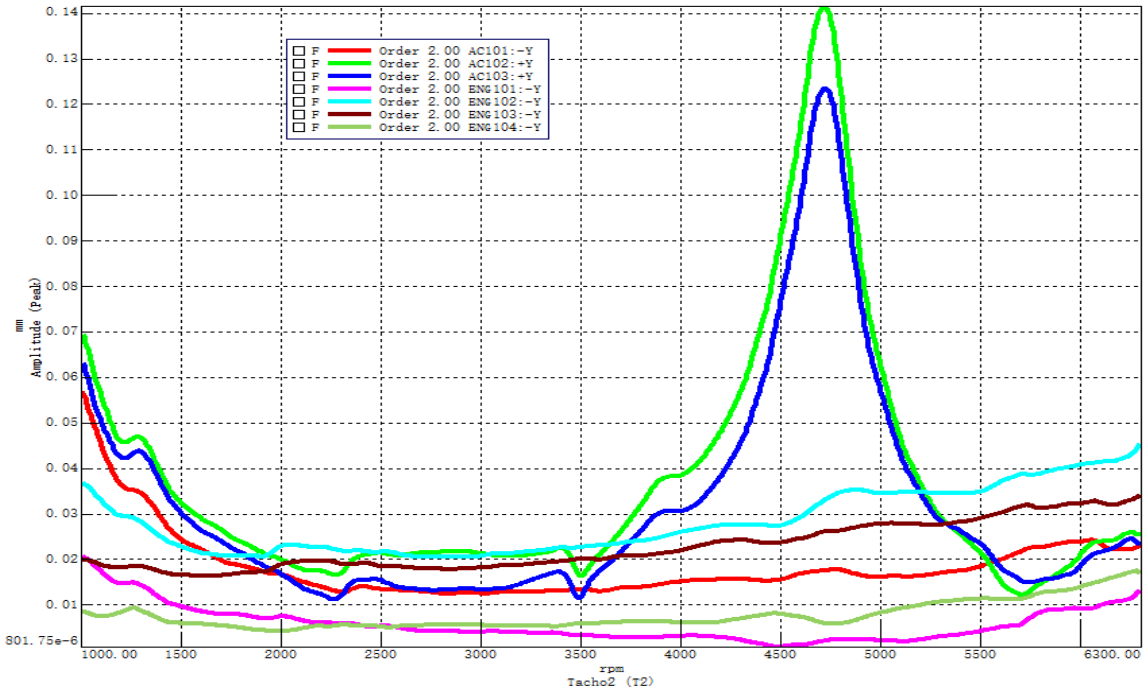


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Displacement_Order_2_X_axis

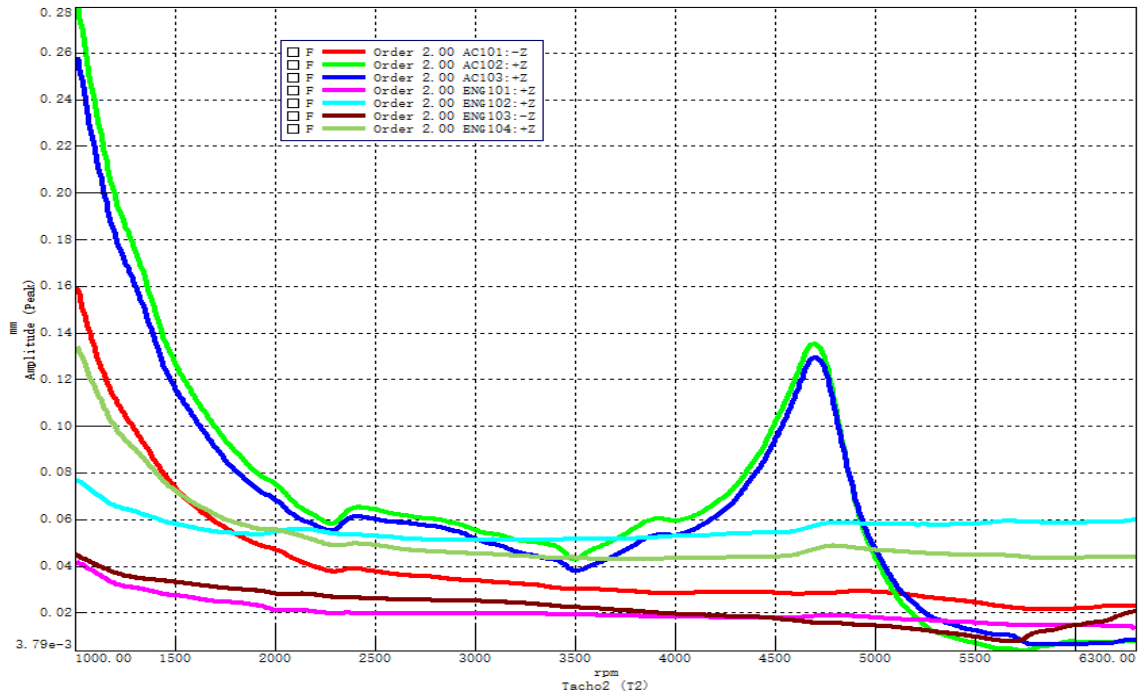


Displacement_Order_2_Y_axis

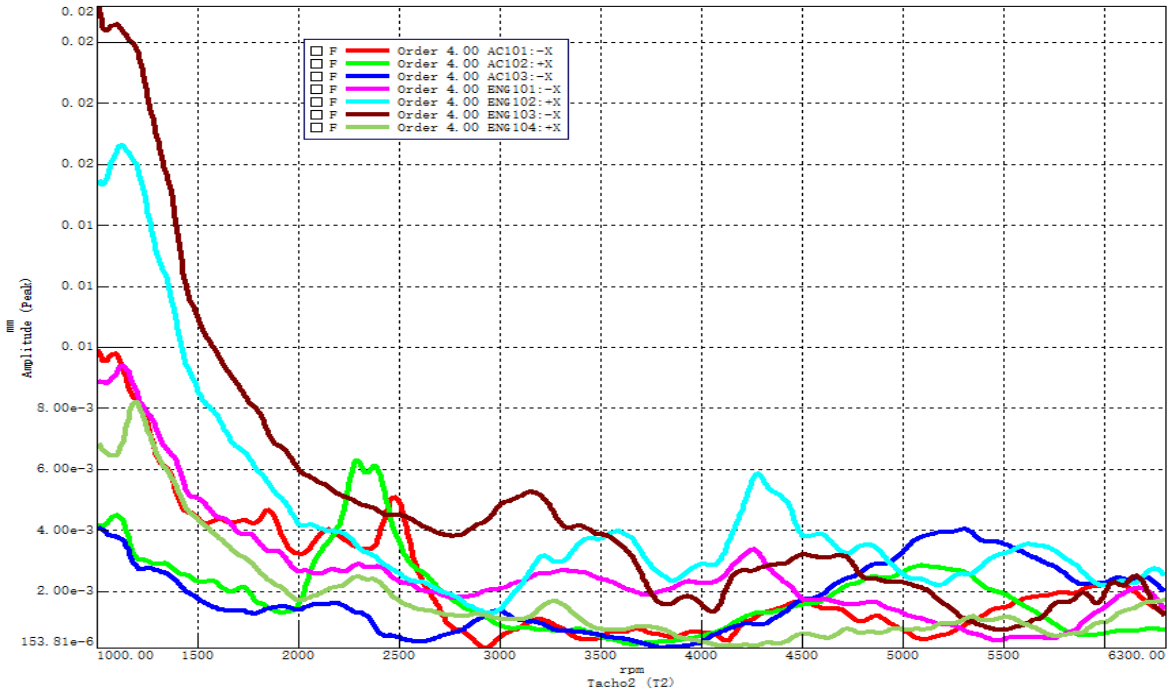


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Displacement_Order_2_Z_axis

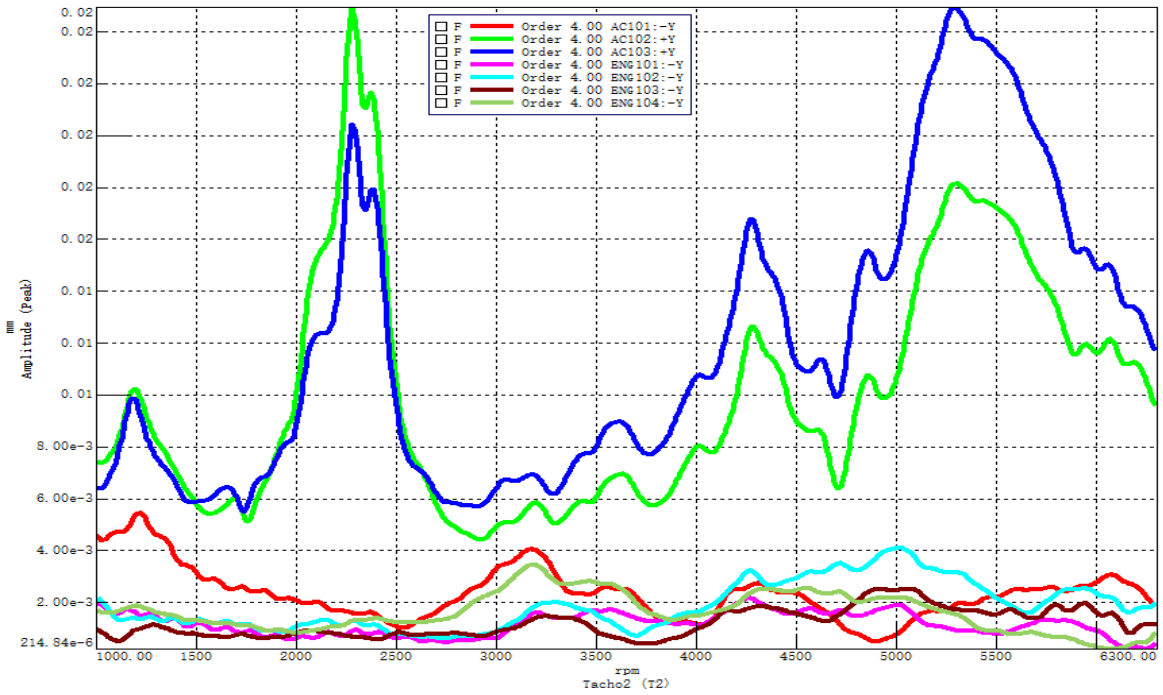


Displacement_Order_4_X_axis

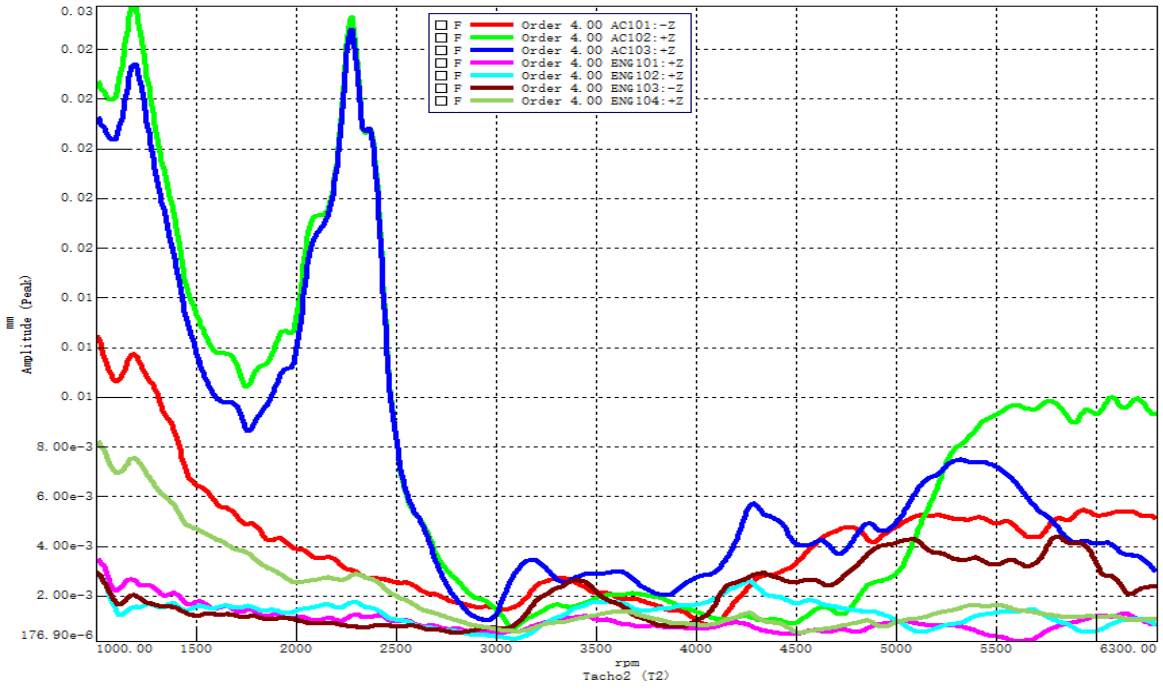


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

Displacement_Order_4_Y_axis

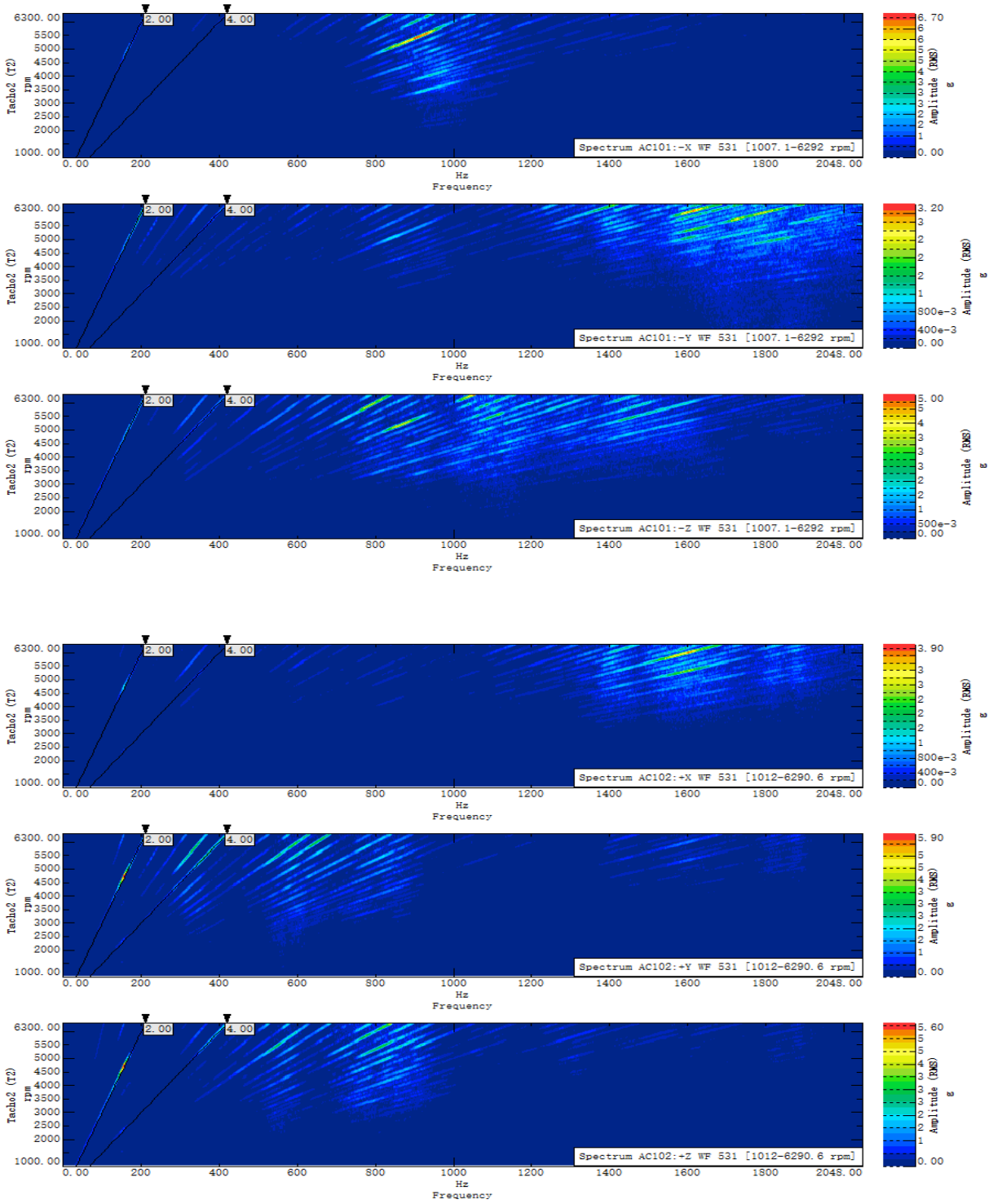


Displacement_Order_4_Z_axis

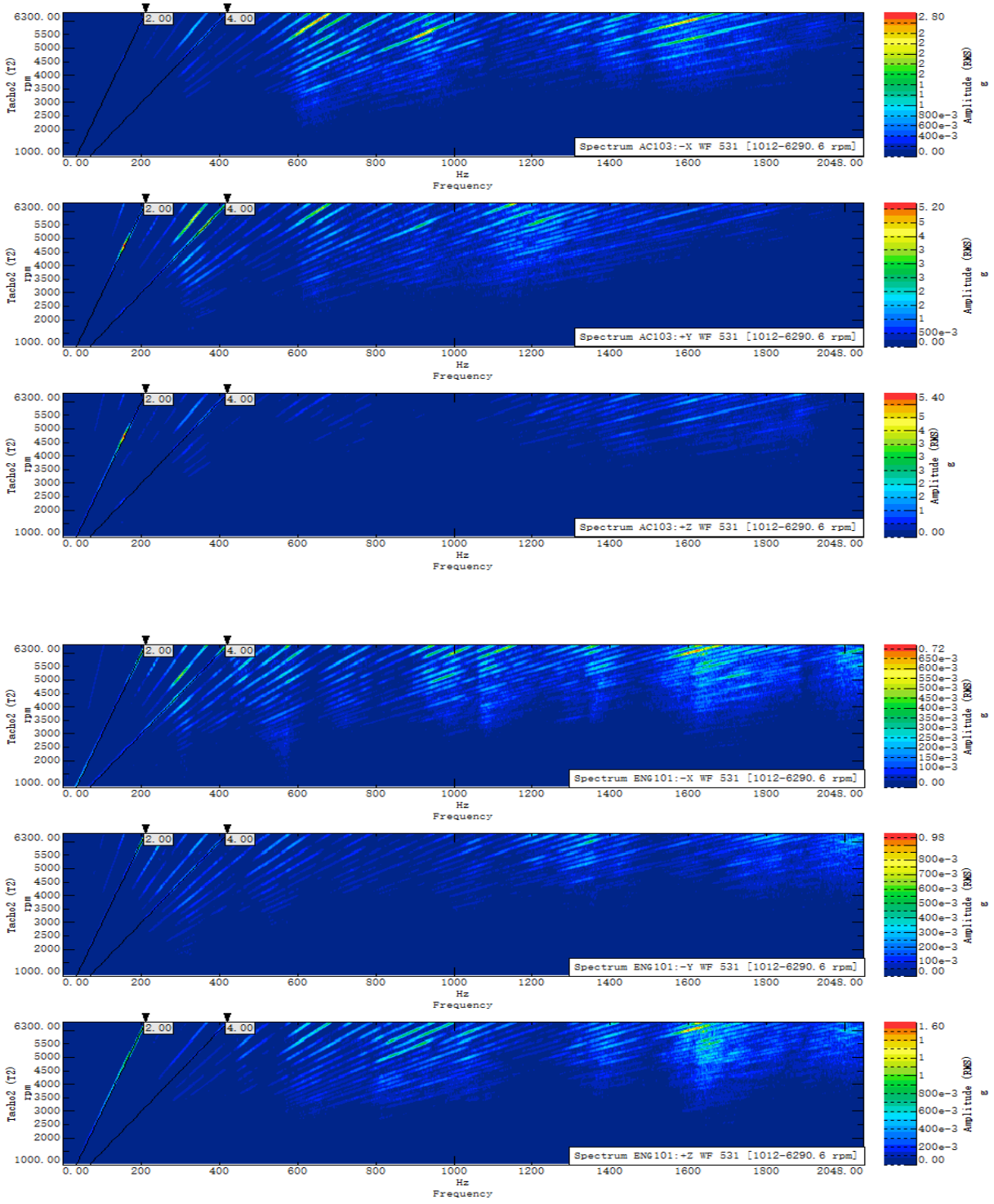


FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

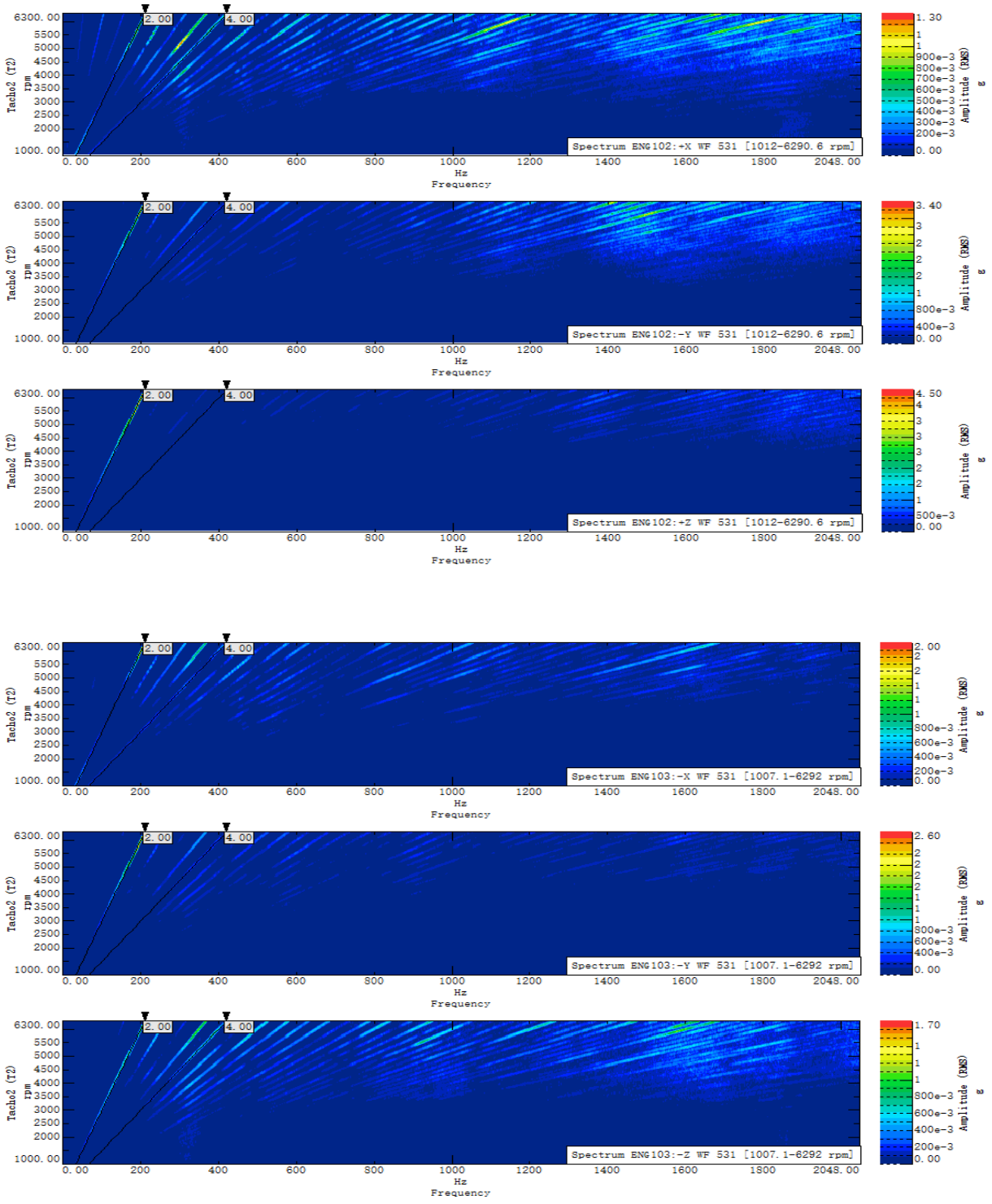
Appendix 4. Vibration measuring



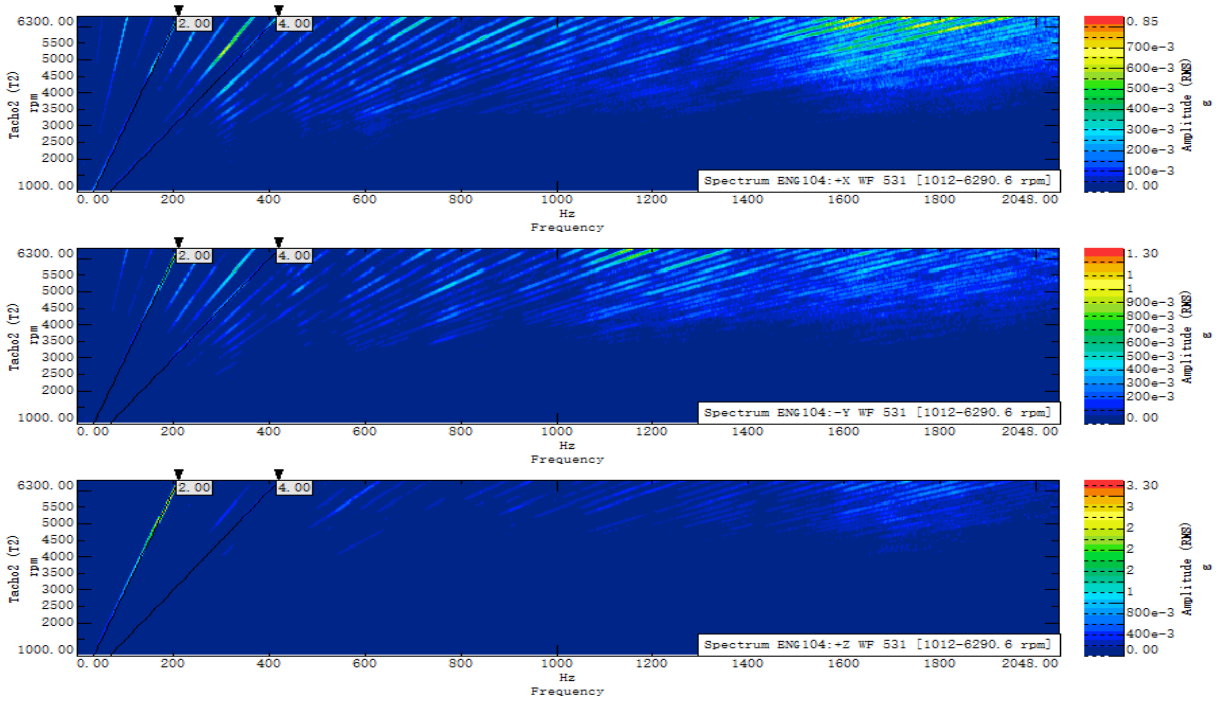
FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040



FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040



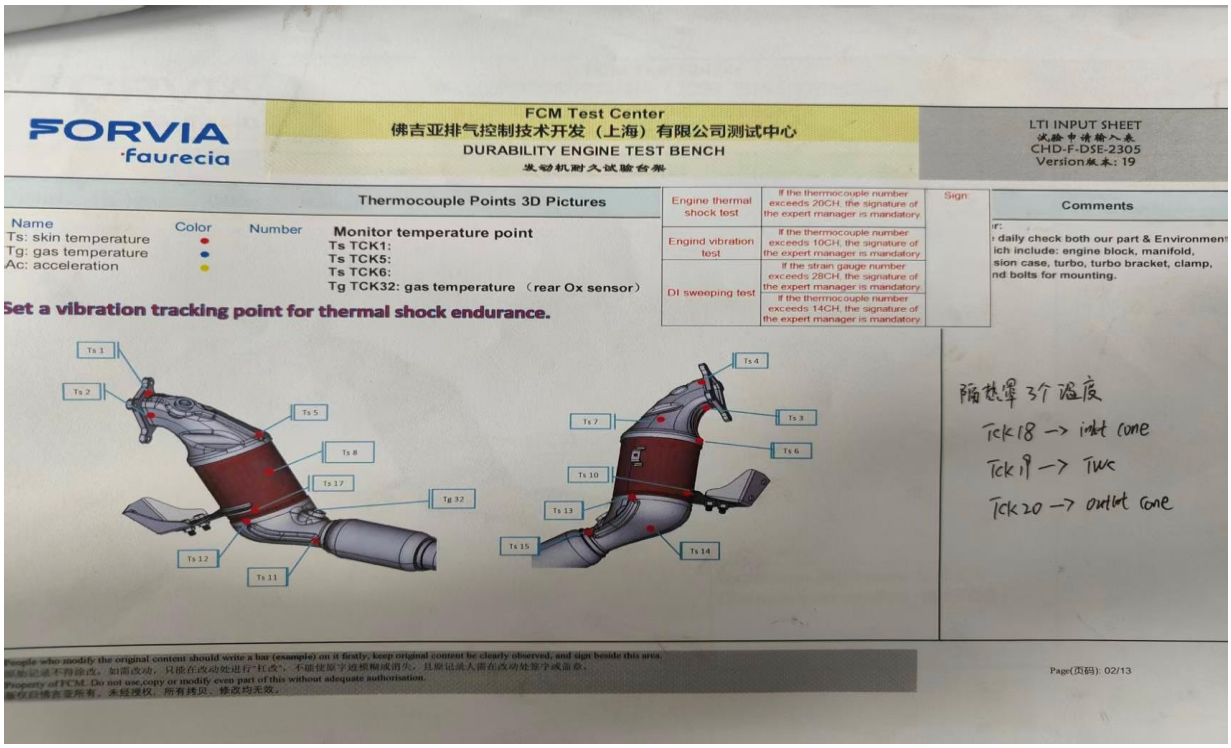
FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040



FORVIA faurecia		LTI_504040

附录 Appendix

5 温度布置 temperature set up



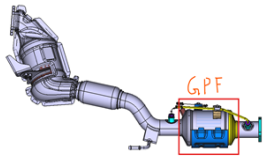
FORVIA faurecia	Engine Durability Sign-Off Sheet	LTI_504040

附录 Appendix

6. 温度测量 Temperature measuring

SPEED	TCK_01	TCK_02	TCK_03	TCK_04	TCK_05	TCK_06	TCK_07	TCK_08	TCK_10	TCK_11	TCK_12	TCK_13	TCK_14	TCK_15	TCK_17	TCK_18	TCK_19	TCK_20	TCK_32
3000	602.6	644.3	618.7	578.1	777.1	746.6	778.4	584.8	608.7	691.2	710.4	737	780.5	660.4	605.3	328.7	250.6	253.5	841.9
3100	604.8	645.9	619.3	581.1	778.7	752.4	778.7	607.3	629.5	694.7	716.6	749	780.2	659.4	619.4	325.5	270.4	251.9	840.2
3200	605.5	646.4	617.2	582.4	775.8	751	776	609.4	632.1	692.1	713.4	747.3	774.4	656.4	621.2	319.9	268.8	251.7	835.2
3300	608.6	647.6	618.9	584.1	775.2	749.9	775	609.1	630.1	688	709.8	744.2	769.7	651.5	617.7	322.3	270.9	248.8	831.7
3400	596.2	636.2	604.7	572	757.5	732.5	756.8	597.4	616.1	668.1	688	723.4	743.9	631	600.9	309.9	260.5	234.9	805.1
3500	596.2	637.7	606.4	574.6	757.4	731.4	756.7	591.6	609.8	665.2	684.3	718.5	740.9	632.1	595.9	307.7	256.1	237.7	803.8
3600	599.1	639.7	606.4	577.1	757.8	731.5	757.3	588.5	607.2	663.1	682.2	717.3	740	634	593	310.8	254.1	233.8	803.5
3700	601.4	641.9	609.4	579.3	758.1	732.2	757.8	587.1	607.2	665.4	682.8	717	739.6	634.3	594.5	310.8	254.7	236.3	803
3800	603.2	643.7	610.5	581.8	758.5	732.6	758.4	586.7	606.5	666	683.4	717.2	739.5	637.2	595.5	311.7	251.5	233.1	802.6
3900	605.3	645.8	611.6	583.2	760.6	733.2	759	587.5	607.7	669.3	686	717.9	740	635.1	598.2	309.4	254.8	232.8	802.9
4000	605.8	646.7	611.6	585.6	761.5	733.7	759.4	586.4	607.7	668.5	686.1	718.7	740	641.5	597.9	308.8	252.6	234.8	802.7
4100	610.3	649	615.6	588.6	761.5	735.6	760.8	586.8	608.6	671.1	687.8	720.5	741.4	642.5	600.2	315	257.9	238.2	803.5
4200	610.7	650.5	616.2	590.5	764.7	736.9	761.2	588.2	610.1	674.1	690.5	721.7	742.2	642.8	602.2	315	253.6	235.8	804.2
4300	614.8	652.9	618.1	594	764.4	738.5	762.6	587.8	611.3	677	692.5	723.1	743.5	647.2	605	314.6	251.7	234.3	805
4400	617	656.3	621.4	597	768.3	740.9	764.9	590.3	613.6	680.3	695.7	725.7	745.8	650.5	606.9	316.3	256.4	240.8	806.9
4500	618.9	656.8	621.4	599.5	769.8	741.8	765.4	590.5	614.9	682.2	697.4	727	746.9	652.3	608.2	315	252.8	237.5	807.5
4600	623.3	660.3	625	602.8	771	745.3	768.7	591.9	617.2	686.1	701.7	730	750.4	654.6	611.7	320.2	256.7	240.4	810.9
4700	626.7	664.8	627.4	607.7	776.1	748.9	772.3	594.6	620	689.1	705.3	733.7	754.4	659.8	614.5	320	264.7	244.3	814.5
4800	630	667	630.7	610	777.6	751.2	774.4	595.8	621.4	693.8	708.7	735.4	756.2	661	618	314.8	257.4	239.9	816.3
4900	632.6	670.4	632	613.2	780.9	753.5	776.3	597.4	623.7	697.1	711.7	737	757.8	662	620.4	319.2	260.2	238.9	817.8
5000	634.1	672.3	633.4	615.6	780.2	754.9	777.5	598	625.9	698.6	712.9	738.9	759.2	668.5	621.3	320.1	264.1	244.1	819
5100	636.8	674.9	635.2	619.3	785	756.5	779.5	600.3	626.4	700.9	714.9	740.5	760.7	671.3	622.8	322.2	263.3	245.3	820.9
5200	641.2	678	638	620.6	784.4	758.1	780.8	600.2	628.3	702.7	716.4	741.4	761.7	668.1	625.2	325	263.6	244.9	822.1
5300	643.7	679.8	636.4	622.4	785.2	758.5	781.2	600.4	628.9	704.9	717.9	741.9	761.9	672.1	626.7	324.8	266.8	246.4	822.4
5400	645.3	680.3	635.9	623.8	783.9	758.6	781	600	628.9	704.4	717.2	742	761.8	675.8	625.3	323.2	267.3	248.7	821.8
5500	645.7	679.9	635.1	624.8	783.9	758.5	781.3	600.6	629.4	705.8	718.3	742.2	761.9	675.2	627.1	325.5	261.3	246.9	821.7
5600	650.2	683.5	639.6	628.5	786.1	761.5	784.5	601.1	631.1	708.6	720.4	745.2	765.3	679.9	628.5	327.1	265.1	252.2	825.5
5700	652.9	686.3	641.2	632	791.7	764.5	787.8	604.2	634.2	711.3	724.3	748.4	768.6	684.6	631.3	330.3	267.4	258.2	828.9
5800	655.6	688.2	643.3	632.9	790.4	765.3	788.3	604.3	634.8	713.5	726.3	749.5	769.4	683.6	633.5	331.8	271.9	256.2	829.1
5900	654.8	688.2	642.9	632.9	789.2	764.7	787.1	604.2	635.4	713.9	726.7	749	768.2	683.8	635	331.1	269.9	251.9	827.5
6000	655.2	687.2	640.6	632.2	788.7	763.4	785.7	603.8	634.9	713.1	725.6	748.5	767.1	683.3	633.9	329.8	271.6	252.1	825.4

FORVIA faurecia	**Engine Durability Sign-Off Sheet**	**LTI_504040**
6.偏差 deviation		
偏差描述 Deviation description	**1. GAC T79振动耐久样件非关键区域有气孔泄漏，泄漏量合格。 2. 热冲击样件载体有细微磕破**	
偏差类别 Deviation category	Sample deviation	
原因 Cause	1. 焊接质量问题. 2.搬运或封装问题	
行动 Action	1. 由试验室对气孔泄漏处处进行补焊。 2. 可直接用于试验，以上两处试验中多观察。	
行动结果 Action Result	1. 气孔处于非关键位置，补焊后不易失效。 2. 几乎不影响试验，低风险。	
照片 Pictures	Hello 金老板， T79-G的发动机耐久试验样件存在如下偏差: 1. 5号样件如下两个位置存在微小气孔，肥皂水可探测到漏气，但总的泄露量仍然为0.00L/Min，并且在非关键区域，故申请偏差由实验室进行补焊后进行发动机振动耐久试验。  2. 2号样件完全无泄露，但前级载体存在细微磕破，无明显裂纹，申请偏差进行thermal 的耐久试验 。  请帮忙确认 谢谢~ Best Regards, Chris QIAN 钱星辉 RE: GAC T79-G 发动机耐久试验偏差申请 JIN Xin 收件人 QIAN Chris; DING Yichen 抄送 FANG Yun; ZHAO Wei (WZO) INTERNAL and PARTNERS 转发该邮件的时间为 7/28/2026 1:06 PM。 好的，我没问题。 然后咱们试验过程中多关注这两个地方吧 **From:** QIAN Chris <chris.qian@forvia.com> **Sent:** 2026年7月28日 13:01 **To:** JIN Xin <xin.jin@forvia.com>; DING Yichen <yichen.ding@forvia.com> **Cc:** FANG Yun <yun.fang@forvia.com>; ZHAO Wei (WZO) <wei.zhao.wzo@forvia.com> **Subject:** RE: GAC T79-G 发动机耐久试验偏差申请	
Date	7/29/2026	

FORVIA faurecia	**Engine Durability Sign-Off Sheet**	**LTI_504040**
6. 偏差 deviation		
Deviation description (From Product Specialist)	**3. 捅穿GPF做热冲击试验**	
Deviation category	Spec. deviation	
Cause	GPF会在试验过程中堵塞，导致发动机功率扭矩异常，试验无法顺利完成。	
Action	GPF进行破坏后做热冲击试验	
Action Result	后级载体对波纹管前段的部分影响较小	
Pictures	RE: 偏差申请：GAC T79 thermal试验 JIN Xin 收件人 QIAN Chris; ZHAO Wei (WZO) 抄送 FANG Yun; ZHOU Kevin; DING Yichen; SHAO Paul INTERNAL and PARTNERS 开始使用全部答复: 好的，谢谢。 谢谢！ 好的。到时候见。 反馈 好的，我没问题 **From:** QIAN Chris <chris.qian@forvia.com> **Sent:** 2026年7月31日 10:43 **To:** ZHAO Wei (WZO) <wei.zhao.wzo@forvia.com>; JIN Xin <xin.jin@forvia.com> **Cc:** FANG Yun <yun.fang@forvia.com>; ZHOU Kevin <kevin.zhou@forvia.com>; DING Yichen <yichen.ding@forvia.com>; SHAO Pa **Subject:** RE: 偏差申请：GAC T79 thermal试验 Hello 老赵， 考虑到后级载体对波纹管前段的部分影响较小，以及以往的项目经验，可以对GPF进行破坏后测试。 破坏后再次确认下外特性的差异。  @JIN Xin 鑫哥，我也和高博确认过，以往类似项目也是按照同样的操作流程进行的。 帮忙也确认下 Hello Chris GAC T79的振动试验即将结束，基于之前GAC类似结构的热端发动机耐久试验的经验，在thermal试验时GPF会积碳堵塞（中后期需2小时左右运行一次再生工况，且有载体高温碎裂的风险），所以我们在振动耐久试验开始前在GPF前加了一个背压监测点，试验刚开始时背压为140mbar左右，现运行了16小时左右，背压上升至340mbar左右，扭矩低了5牛米。所以建议在thermal试验开始前，把GPF摘穿（同P36等项目）后进行试验。 @QIAN Chris请分享一下偏差建议，谢谢! @JIN Xin金总：请确认一下偏差，谢谢!	
Date	7/31/2026	